Rich Internet Applications for Business Applications

How things started

- Coming together of midrange software companies developing business applications / solutions (Oct 2007)
- Result
  - Business processing side is „under control“
    - Java, Spring, Hibernate, Web Services, Jasper, ...
  - Front end side is still „in discussion“
    - **No satisfying strategy to convert the core application**
      - Often: Fat client still alive
      - HTML, AJAX experience avilable with all companies
    - Goal: Rich Client in front of server side application

Challenge: Application Lifecycle

- Application Lifecycle: >> 5 years
  - ==> Lllooonnnnngggg term commitments are made
- Technical Discussion and...
- Management Discussion
  - Build?
  - Buy?

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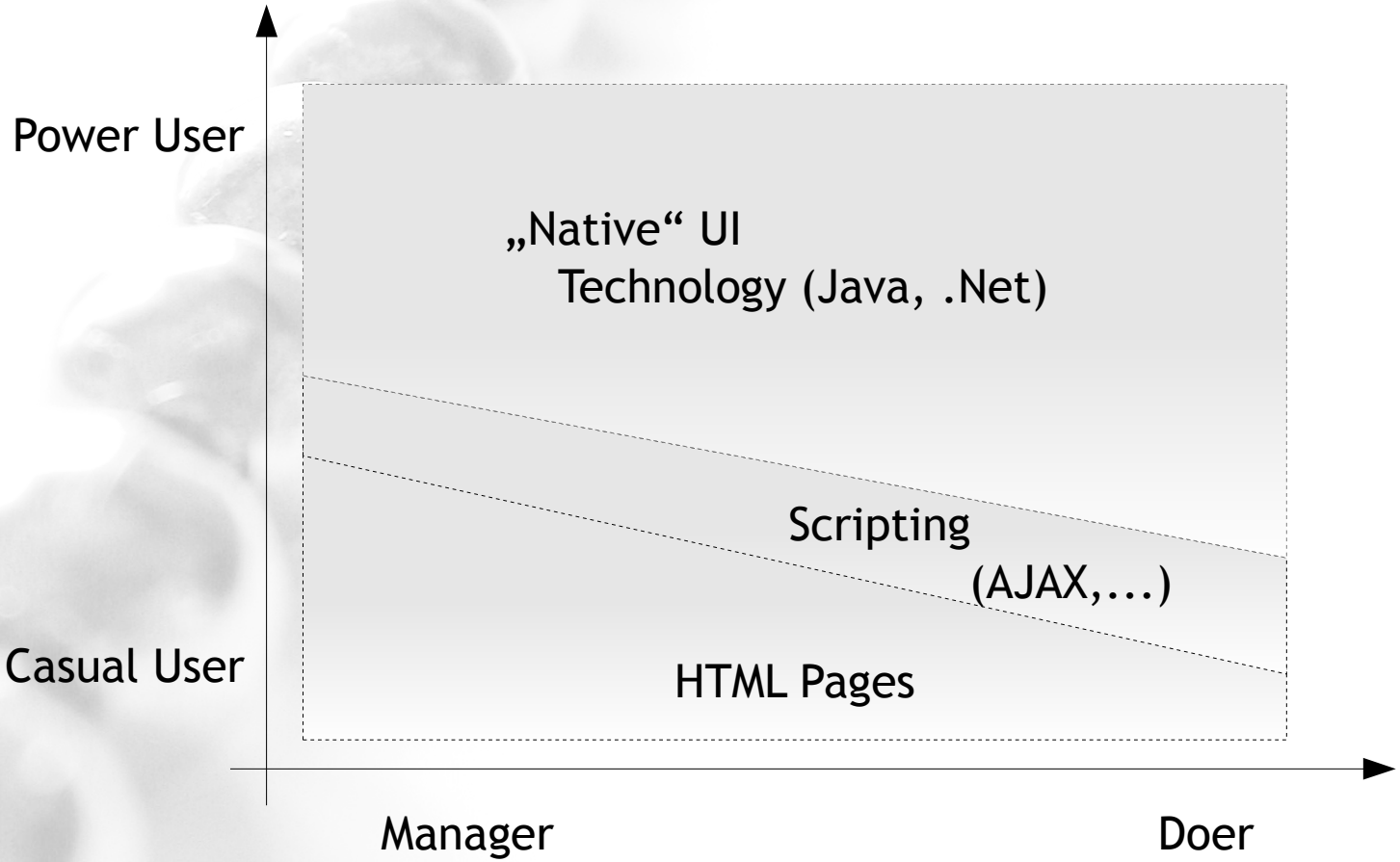
- Technical Discussion and...
- Management Discussion
  - Build?
  - Buy?
  - ...Community!
    - Share vision
    - Share code, problems, knowledge
    - Share risk

User Types and UI Technology



„Typical Power User“

- Quality — Doer!
- Frequence — Frequent!
- Processing — Display / Edit
- Relation — Known

- Examples
  - Call Center Employee
  - Purchasing Manager
  - Financial Accountant
  - Production Planner
  - Developer - IDE

Power User's Expectations

- User Interface needs to be...
- FAST
  - Response times to be measured „in milliseconds“
- ROBUST
  - Whole day usage without restart / slow down
- INTERACTIVE
  - Up to date control processing: drag & drop, popup menus, table sizing, ...
  - Keybooard-able
- SMART
  - Look and feel

„Typical Casual User“

- Quality — Tactical User ... Doer
- Frequence — Sometimes
- Processing — Display ... Edit
- Relation — Unknown

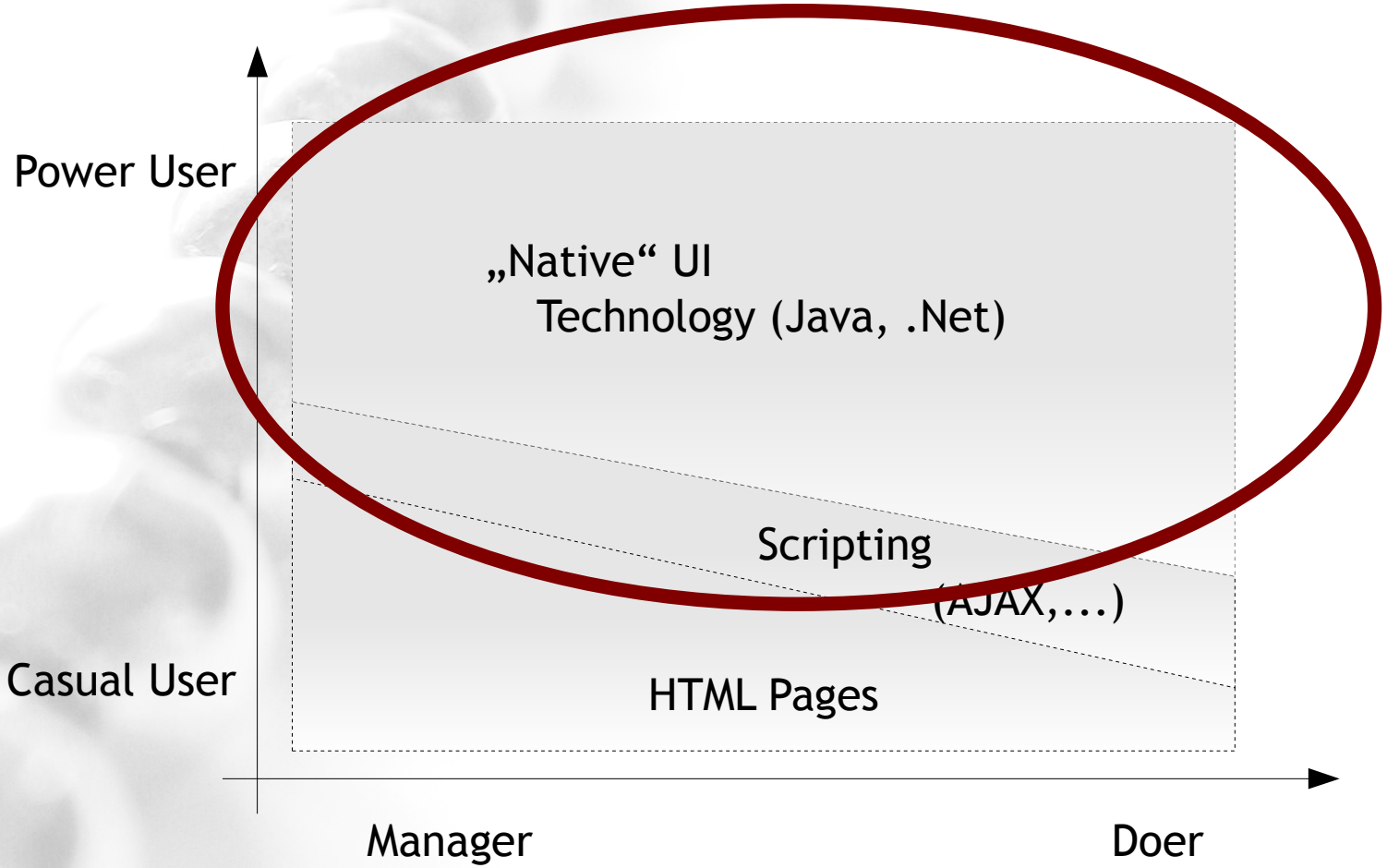
- Examples
  - Web Mail
  - Mini Office User
  - Travel booking
  - Developer - Newsgroup

Casual User's Expectations

- User Interface needs to be...

- AVAILABLE
  - No client installation / configuration
- SMART
  - Look and feel
- INTERACTIVE
  - Avoid „annoyance“
  - Example: calendar input, ...

Operationally used Business Applications

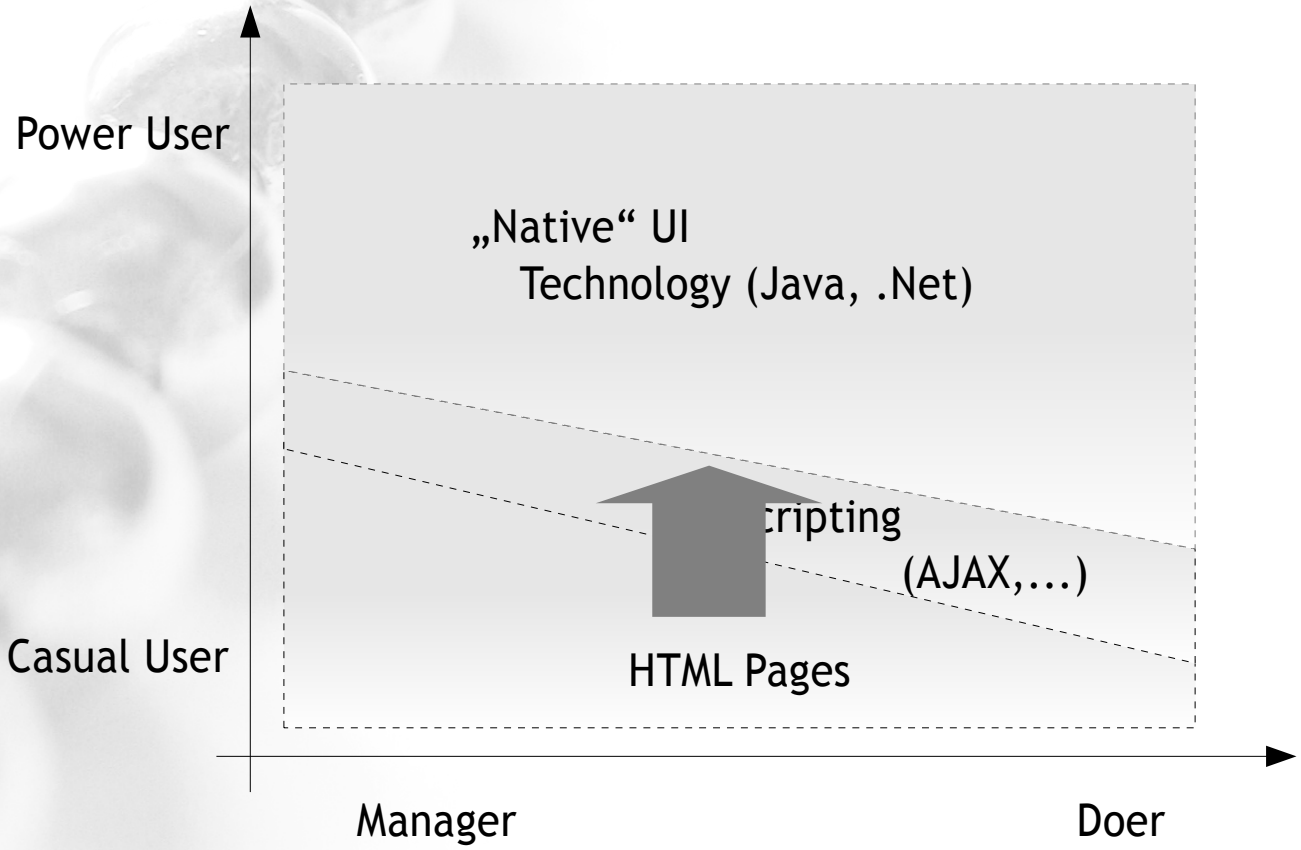


And here comes... AJAX!

- PREJUDICES (page one)
  - AJAX improves the interaction comfort
  - AJAX gives HTML pages a flavour of desktop applications
  - AJAX provides frameworks which manage the complexity of HTML / Javascript / ... processing
  - AJAX is (still) an area of hype, there is a lot of development going on worldwide
  - AJAX is complex, but there are frameworks that hide complexity
- GOOD NEWS: THEY ALL ARE TRUE!

AJAX Expectations

- Lift up the area of HTML browser usage...!

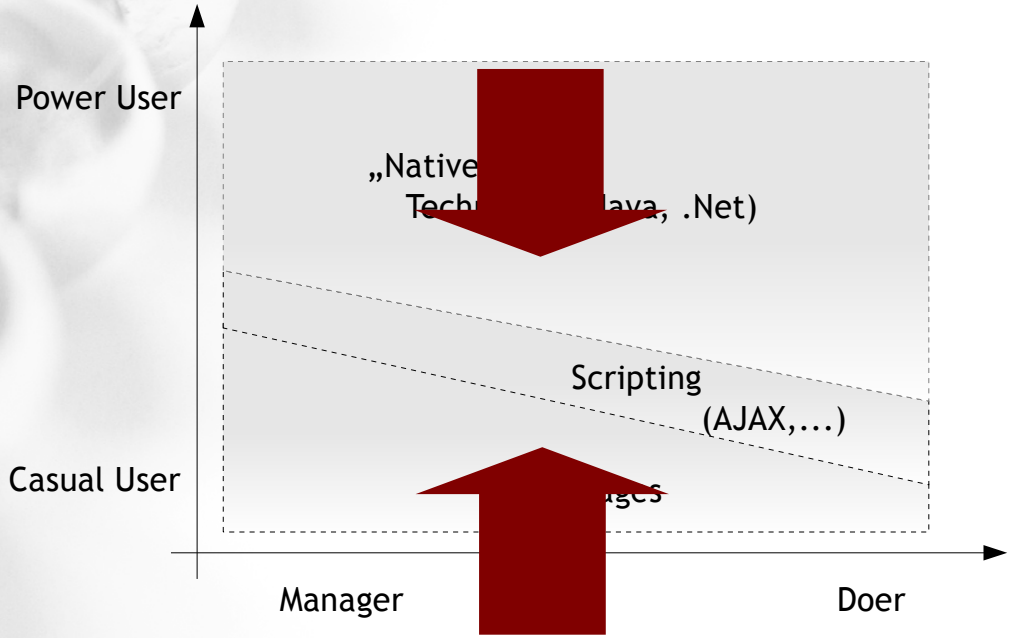


And here comes... AJAX!

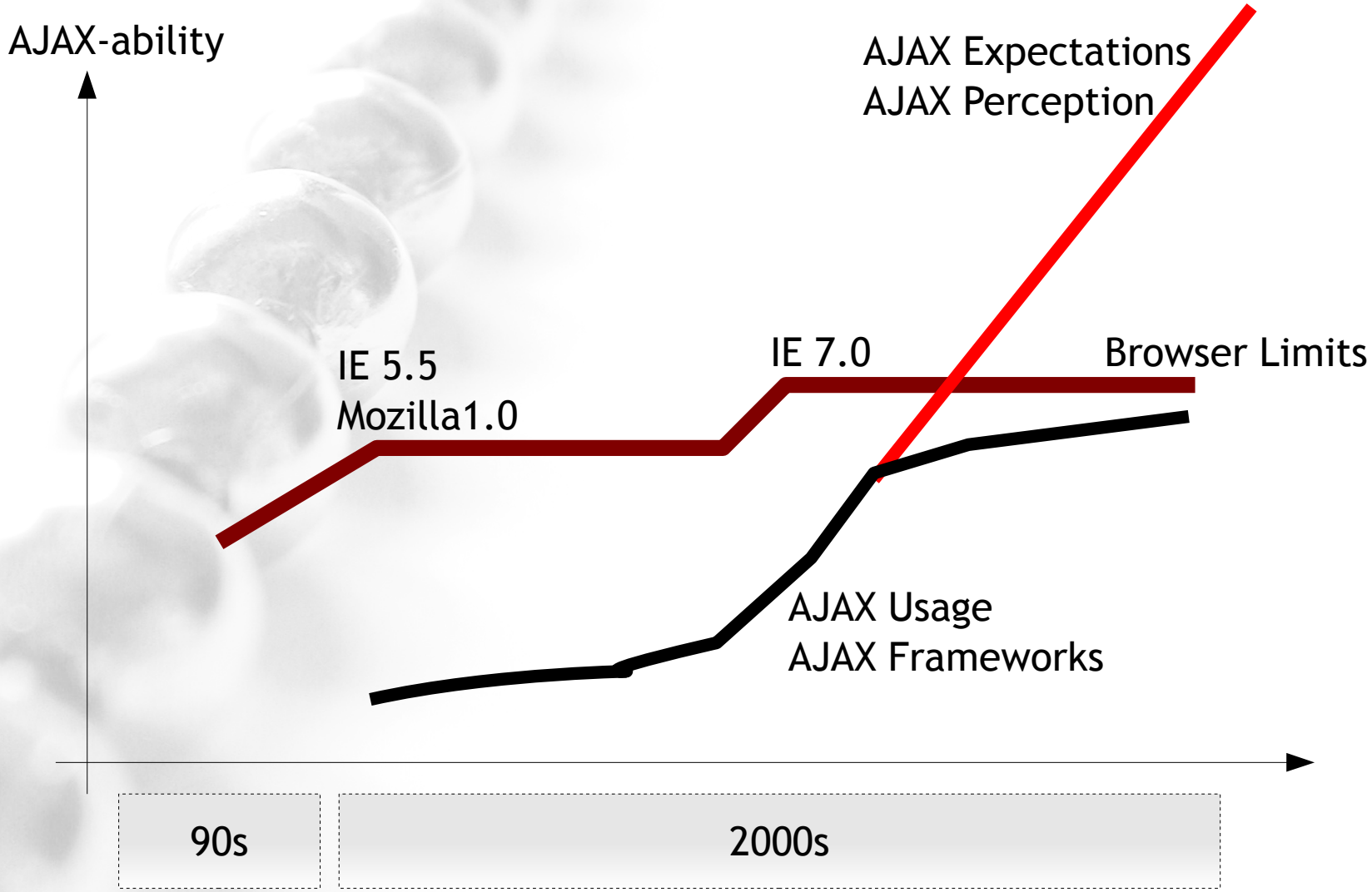
- PREJUDICES (page two)
  - AJAX client rendering is slow
  - AJAX frameworks either work fine with IE or with Mozilla
  - Using AJAX frameworks means you should know the private phone number of the framework developers
  - Browsers are oracles
  - Missing standards
- BAD NEWS: THEY ALL ARE TRUE. Sorry.

AJAX Usage Dilemma

- Casual users want simple screens to be loaded fast!
- Operational users want complex screens to operate fast!



AJAX means „Fighting with Limits“



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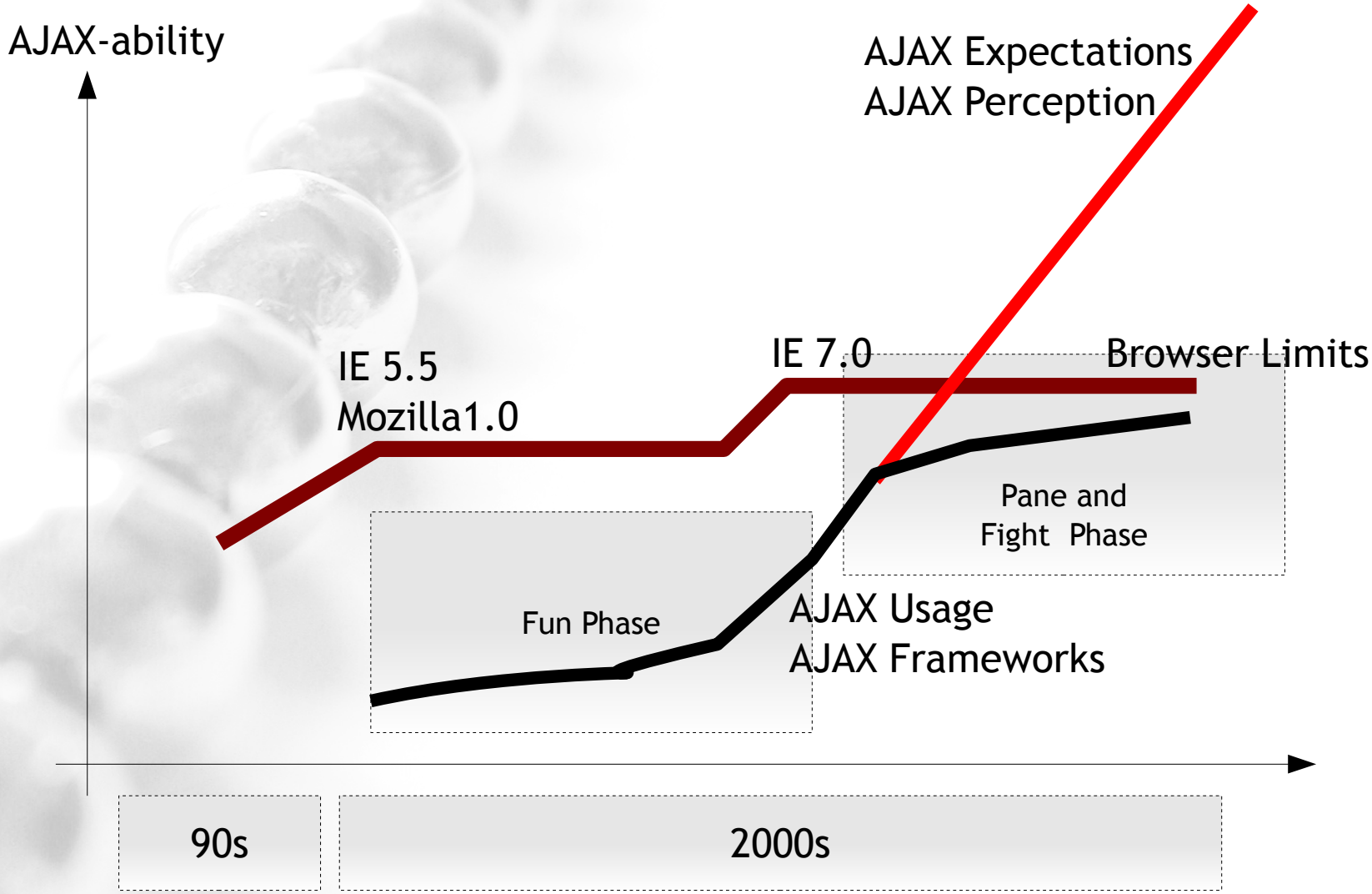

AJAX Maturity Conclusions

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GoogleEarth

GoogleMaps

What UI Technology to choose...

- ...for operationally used Business Applications

- AJAX
  - Performance, Robustness, Compatibility
- Macromedia
  - Animation driven, Valid Option
- MS Silverlight
  - Animation driven, young, Valid Option (Future)

- ..or..

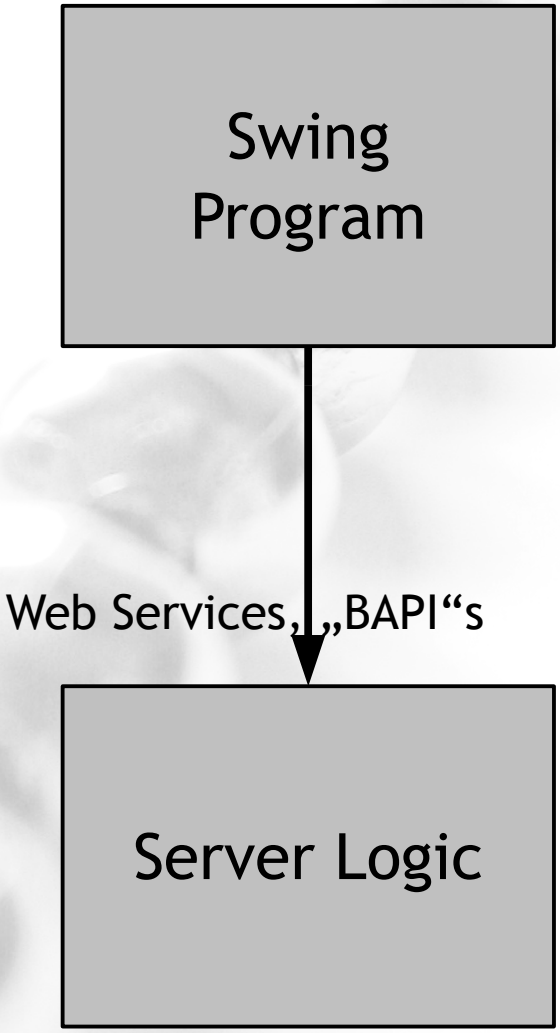
Java Clients

- „Industry proven“
  - Compatible throughout „all“ client OS platforms
  - Fast
  - Robust
  - Used in many, many applications
- Swing vs. Eclipse SWT
  - „=“ Performance
  - „+ Swing“ Compatibility
  - Interoperability Swing <==> SWT
- Active area of development
  - Java 6.10 improvements

Java Swing

- Swing...!
  - Available on „all“ platforms - „code once“ is reality
  - Fast and robust
  - Component libraries available
    - Transparency, 2D, 3D, charts, ...
- But...:
  - Swing is just a library – not a rich client concept
  - Swing is tough for application developers
    - Layout Managers...
    - Complexity of UI libraries is usually underestimated
  - Swing looks old fashionned by default

NOT Wanted



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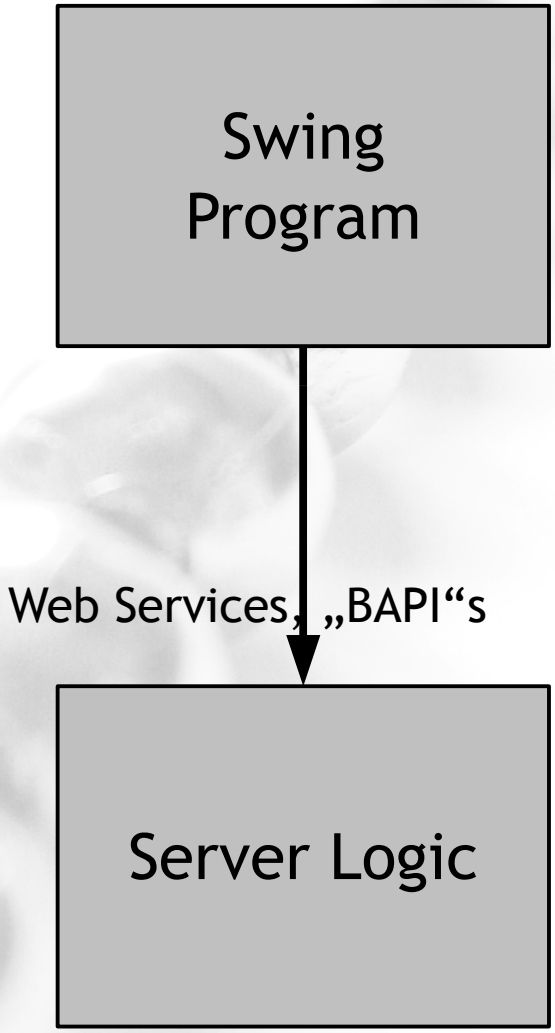
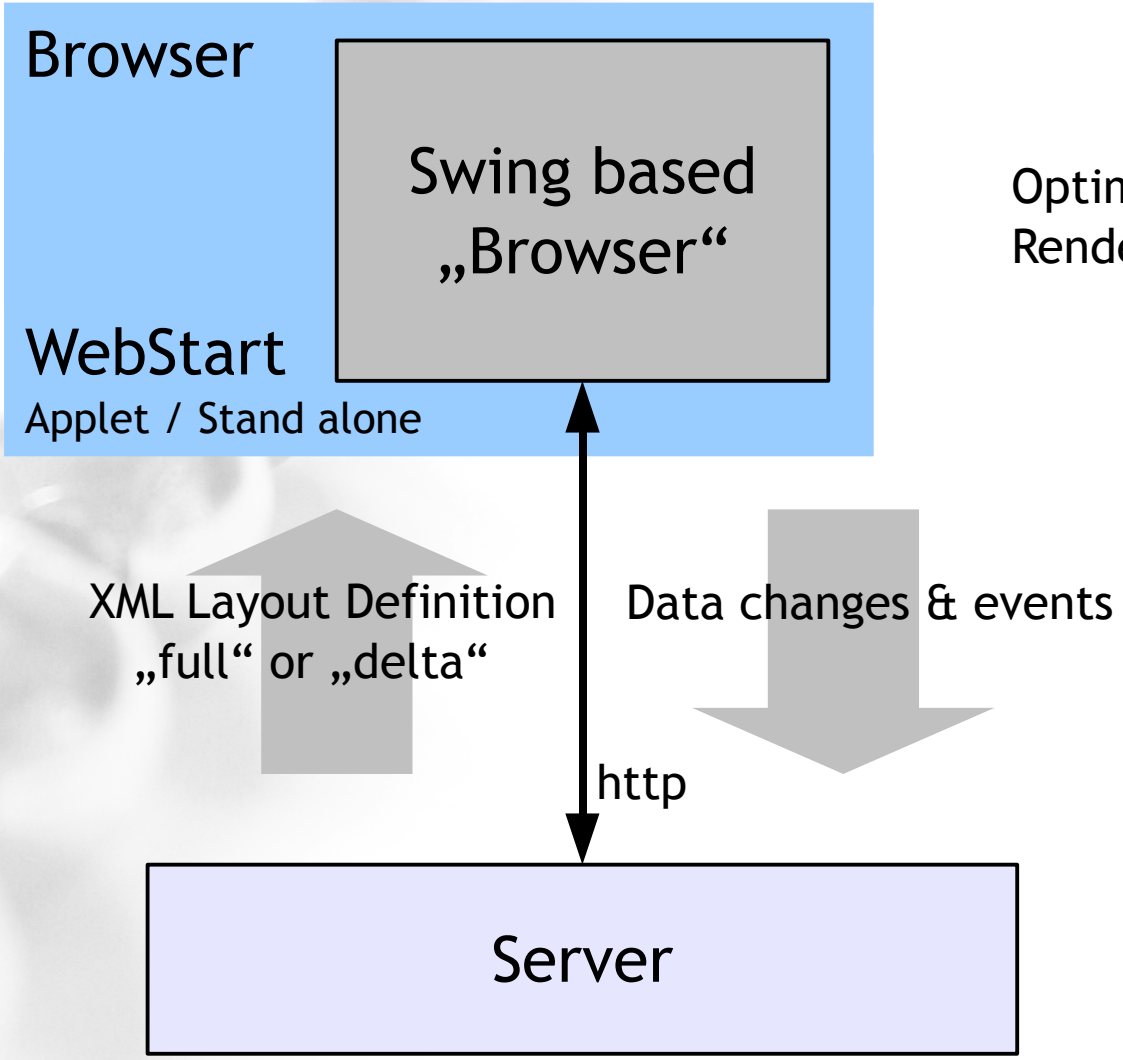


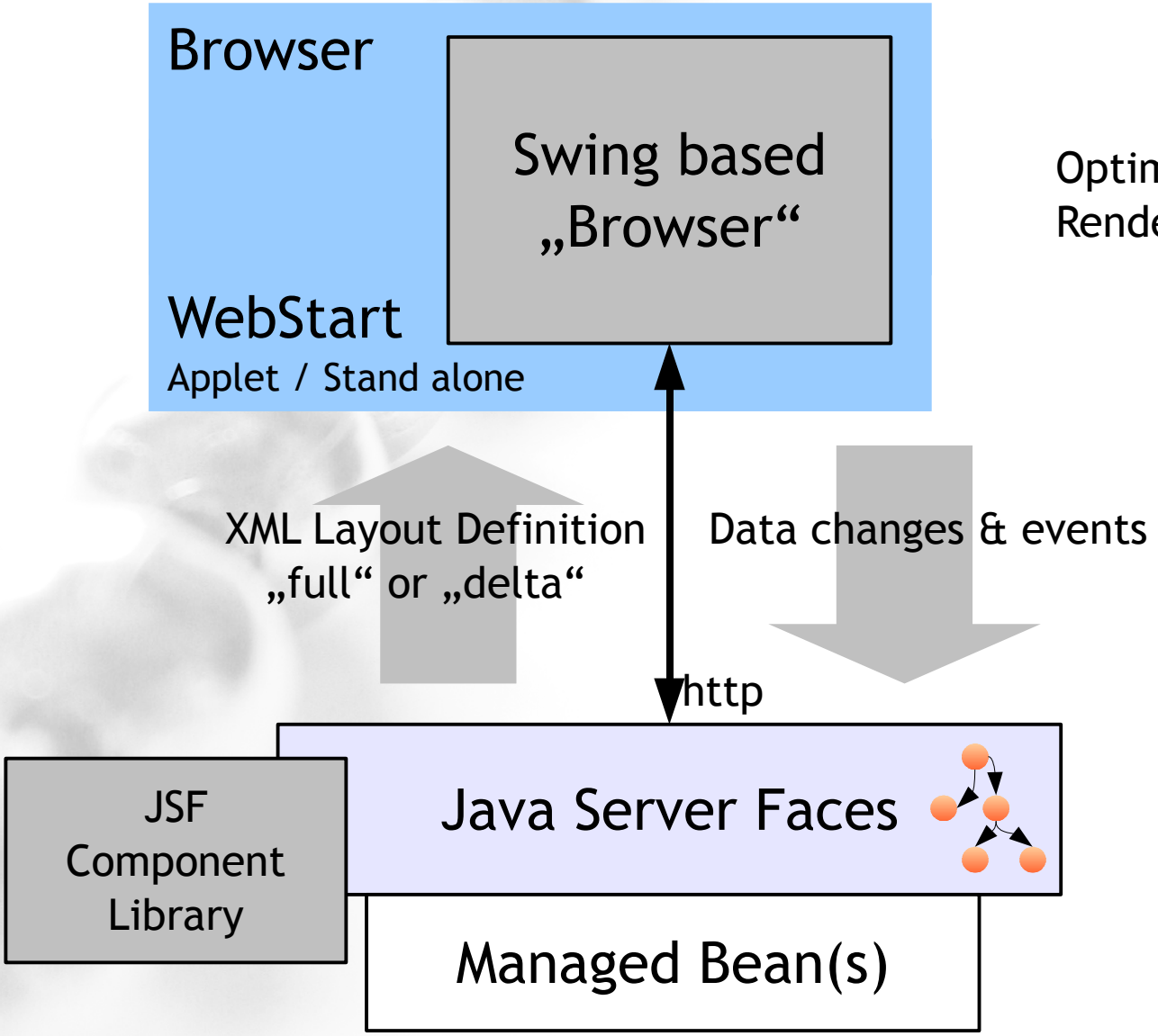
- Development Effort – both frontend and backend development
- WAN-ability – not ensured
- Security – all „BAPI“s to be secured
- Software Distribution – massive client updates

Rich Client Concept



Optimized for delta Rendering

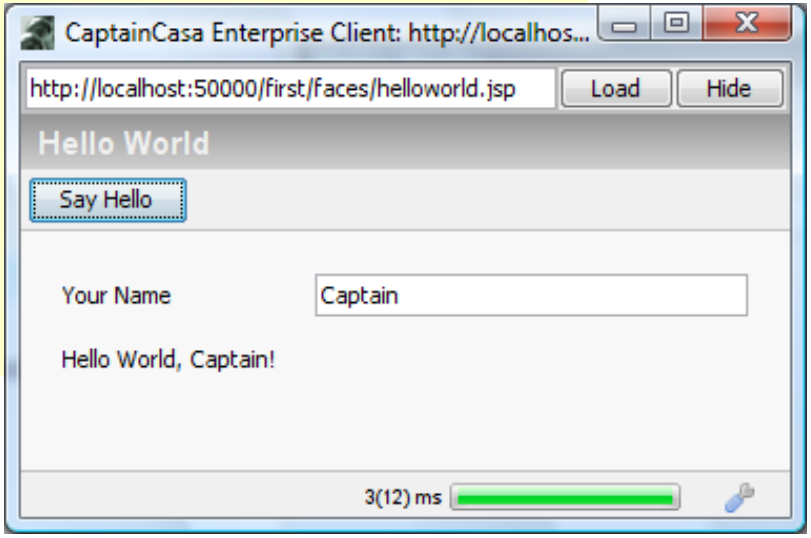
Rich Client Concept



Optimized for delta Rendering

Hello World! - JSP

```
<t:rowtitlebar text="Hello World" />
<t:rowheader>
  <t:button actionListener="#{helloworld.onHello}" text="Say Hello" />
</t:rowheader>
<t:rowbodypane>
  <t:row>
    <t:label text="Your Name" width="120" />
    <t:field text="#{helloworld.yourName}" width="100%" />
  </t:row>
  <t:rowdistance height="20" />
  <t:row>
    <t:label text="#{helloworld.outputInfo}"
                 width="100%" />
  </t:row>
</t:rowbodypane>
<t:rowstatusbar/>
```



Hello World! - Java (Server side)

```
<t:rowheader>
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</t:rowheader>
<t:rowbodypane>
  <t:row>
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    <t:field text="#{helloworld.yourName}" width="100%" />
  </t:row>
  <t:rowdistance height="20" />
  <t:r
    <t:
  </t:
</t:r
<t:ro
```

```
public class HelloWorldBean implements Serializable
{
    String m_yourName;
    String m_outputInfo;

    // Property yourName
    public String getYourName() { return m_yourName; }
    public void setYourName(String yourName) { m_yourName = yourName; }

    // Property outputInfo
    public String getOutputInfo() { return m_outputInfo; }

    public void onHello(ActionEvent event)
    {
        m_outputInfo = "Hello World, " + m_yourName + "!";
    }
}
```

Key architectural Advantages

- Fast, robust, approved Client Technology
  - 50 times faster than HTML / AJAX
- Standard Server Technology
  - Scalability, Failover, ...
- Simple and fast Development
  - Descriptive UI definition
  - Client-Server Separation built in
    - No double-coding (client/server)
  - I18N, ... built in
- Efficient Client Deployment

Conclusion (I)

- Your boss asks you to provide a frontend that...
  - ...runs everywhere in all browsers with zero installation
  - ...is fast and meets requirements of power users
  - ...covers complex screens with „tons of data“
- Tell him/her about problems
  - AJAX/HTML is fine for many scenarios but does not really meet requirements of power users
- Growing role of browser plugins that utilize client UI resources more effectively than HTML/JavaScript

Conclusion (II)

- Selecting „native“ technology is a valid option
  - Intranet, managed environment
  - Known Users
- When selecting a technology
  - Do not build Fat Clients (if not explicitly wanted)
  - Check usage of a generic client renderer
  - Check if you can plug in into exsiting server environments

THANKS!